WEATHER FORECASTING

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Global Weather Trend Forecasting

137K+
Rows

211
Countries

41
Features

3 ML
models

Dataset Overview

- Source: Kaggle — Global Weather Repository
- Date Range: May 2024 → April 2026
- Coverage: 211 Countries, 257 Cities
- Key Features: Temperature, Humidity, Wind, Precipitation, Air Quality (PM2.5, CO, Ozone), UV Index, Visibility

DATA CLEANING AND PRE-PROCESSING

- Zero missing values — no imputation needed
- Dropped redundant columns (Fahrenheit, mph, inches)
- Parsed last_updated as datetime, sorted chronologically
- Engineered new features: hour, month, season, day_of_week

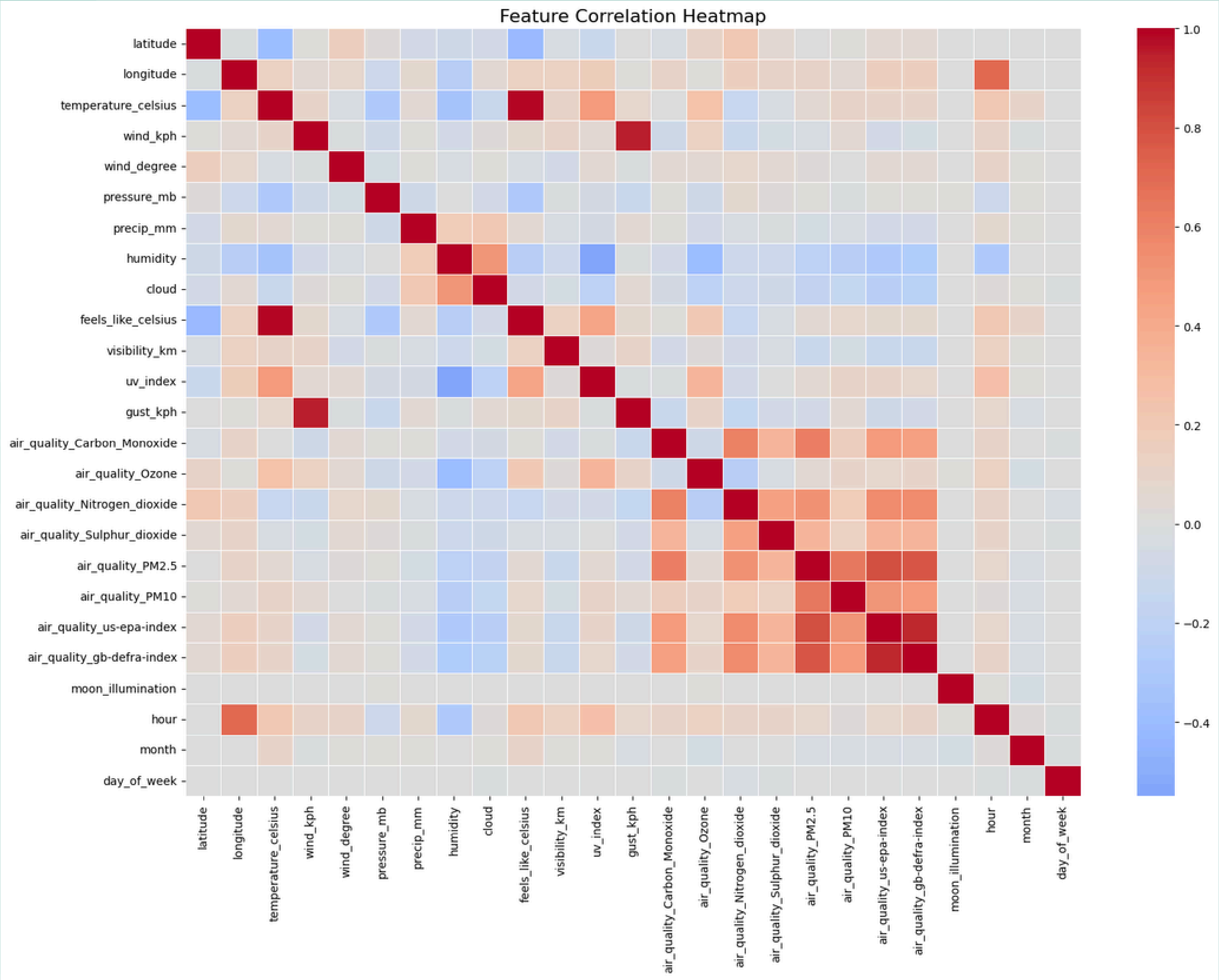
EDA : TEMPERATURE

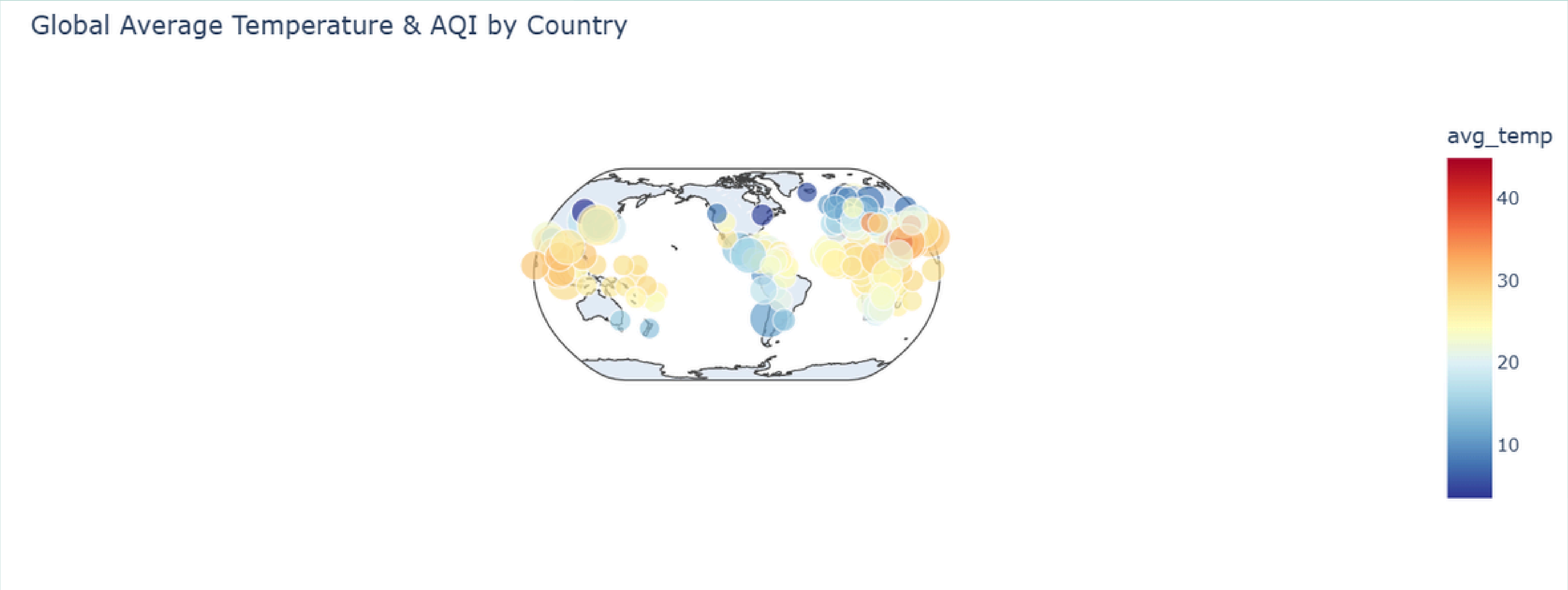
Seasonal temperature distribution across 211 countries

Insight:

Summer average~28 °C
Winter average~8 °C
Clear seasonal separation globally

EDA : CORRELATION AND SPATIAL





Insight:

Temperature & feels like: 0.99 correlation

Equatorial countries hottest

South Asia worst AQI

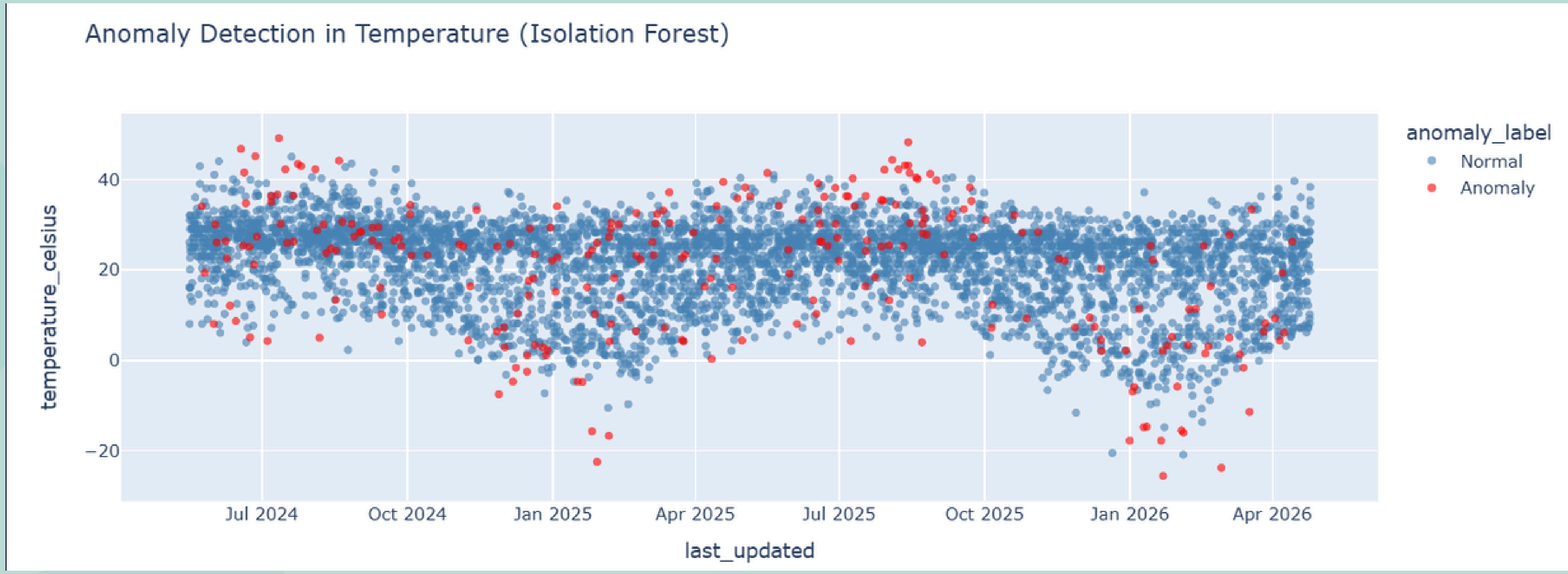
ANOMALY DETECTION

Algorithm: Isolation Forest

Contamination: 5%

Anomalies found: ~6,871 records

<u>These represent:</u> heatwaves, pressure drops, pollution spikes



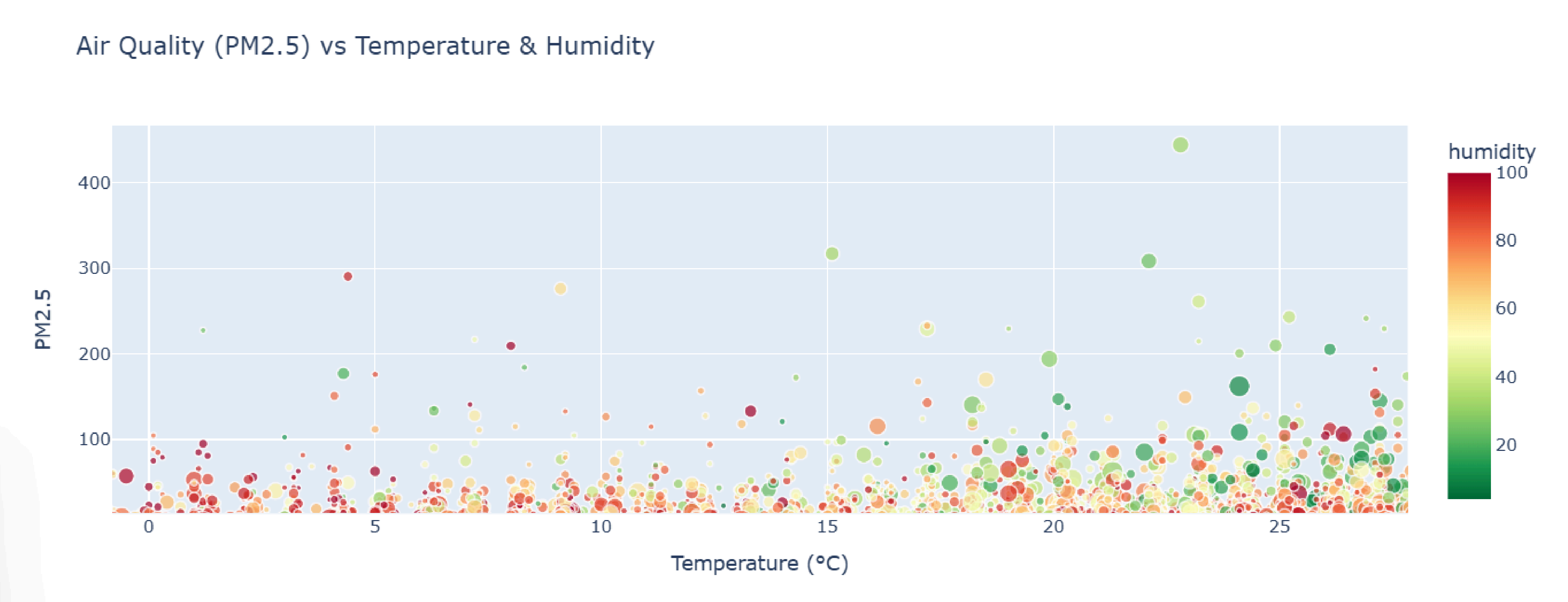
ENVIRONMENTAL IMPACT

Higher temperature → higher PM2.5 levels

High humidity traps pollutants near ground

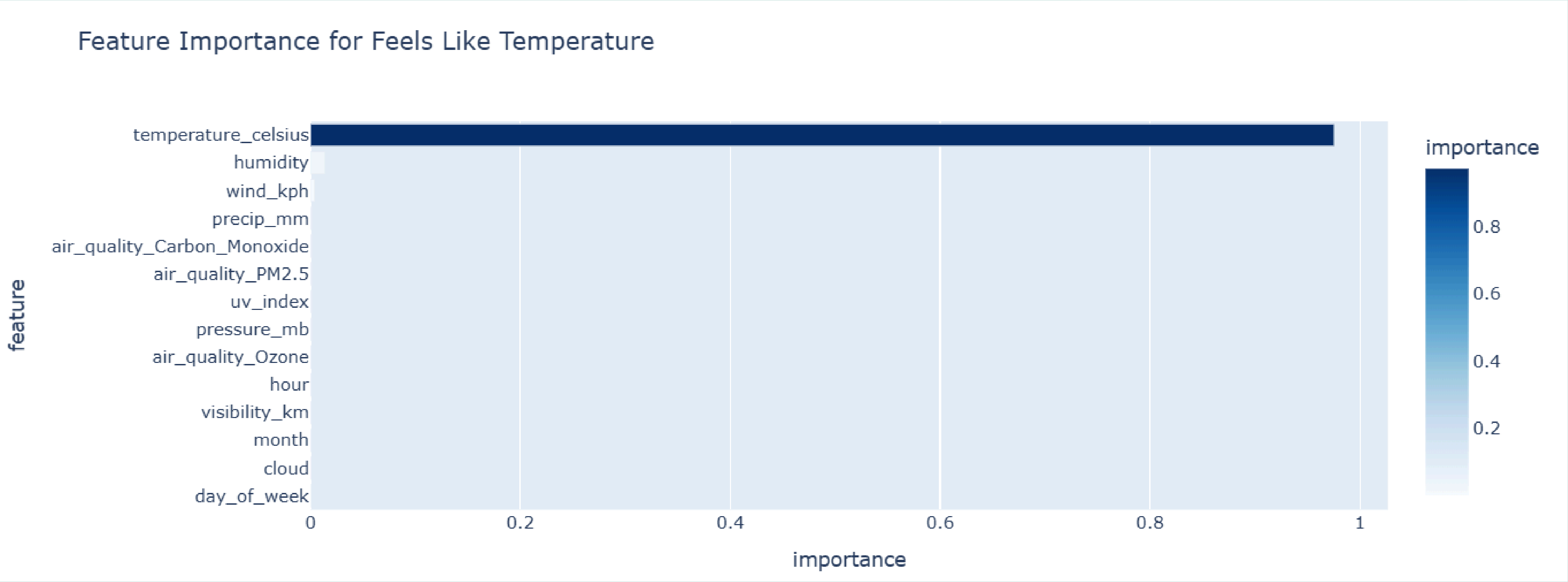
Wind speed >20 km/h reduces PM2.5 significantly

South Asia & East Asia — worst air quality globally



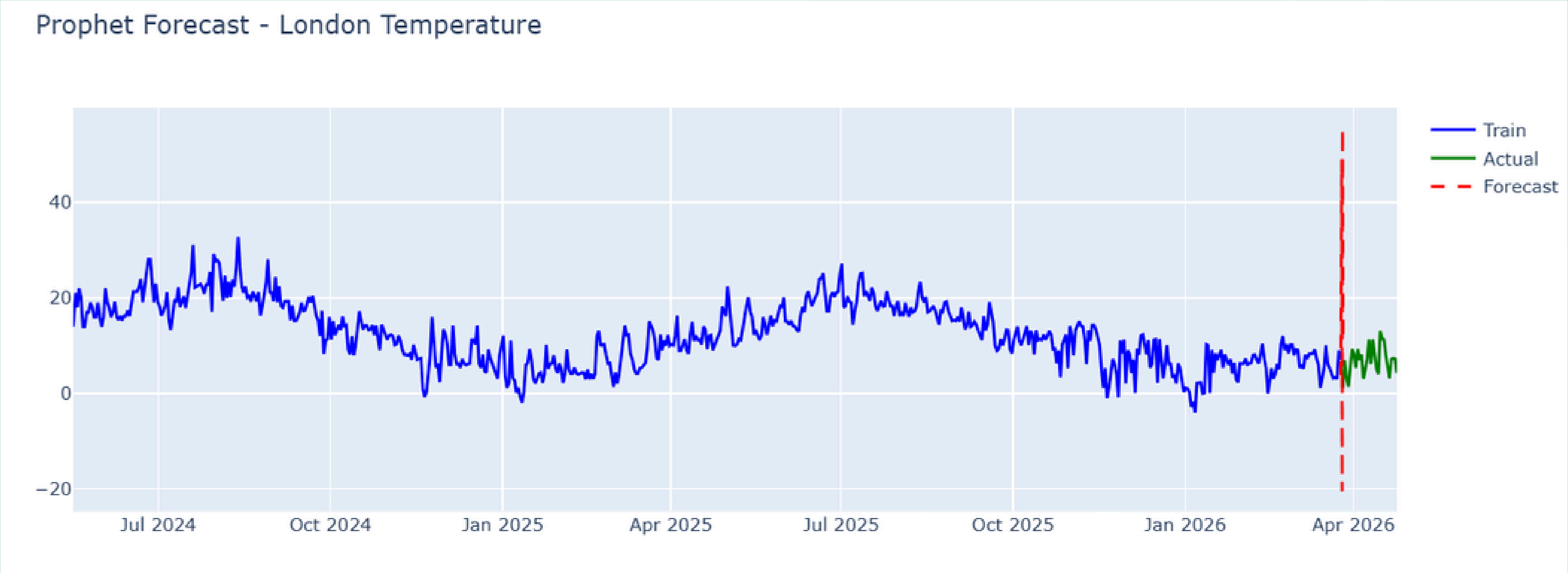
FEATURE IMPORTANCE

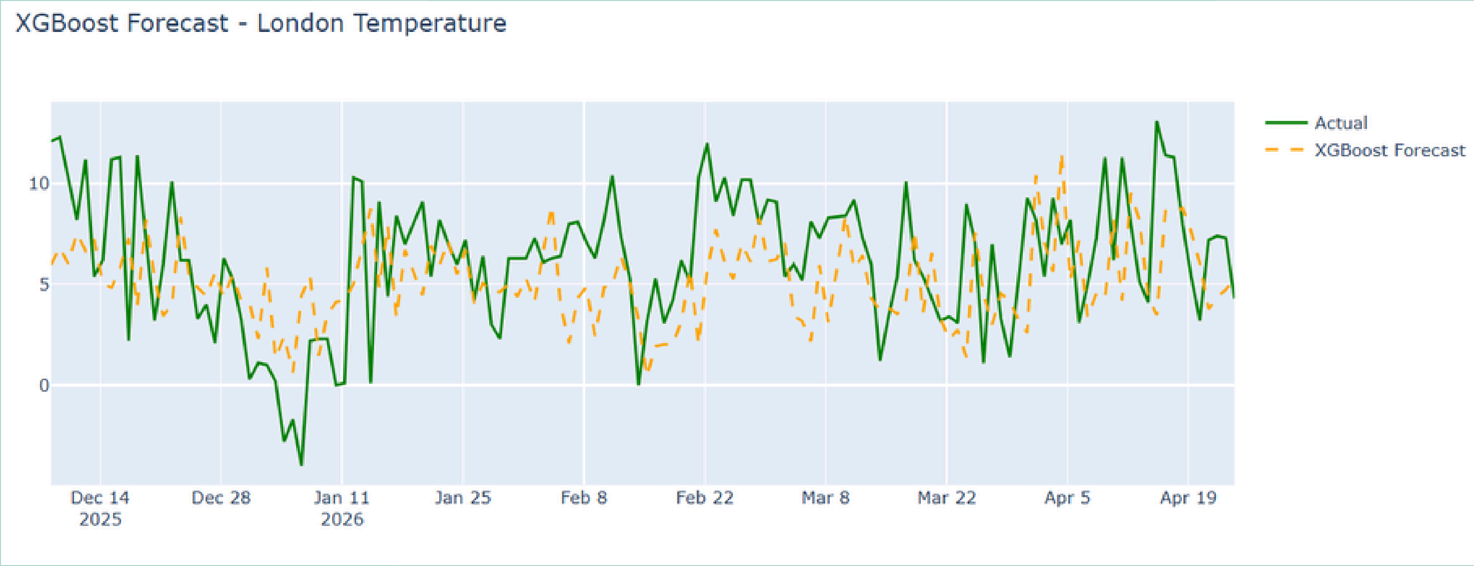
Temperature_celsius: 72%
Humidity: 9%
Wind_kph: 7%
Pressure_mb: 4%
Cloud: 3%

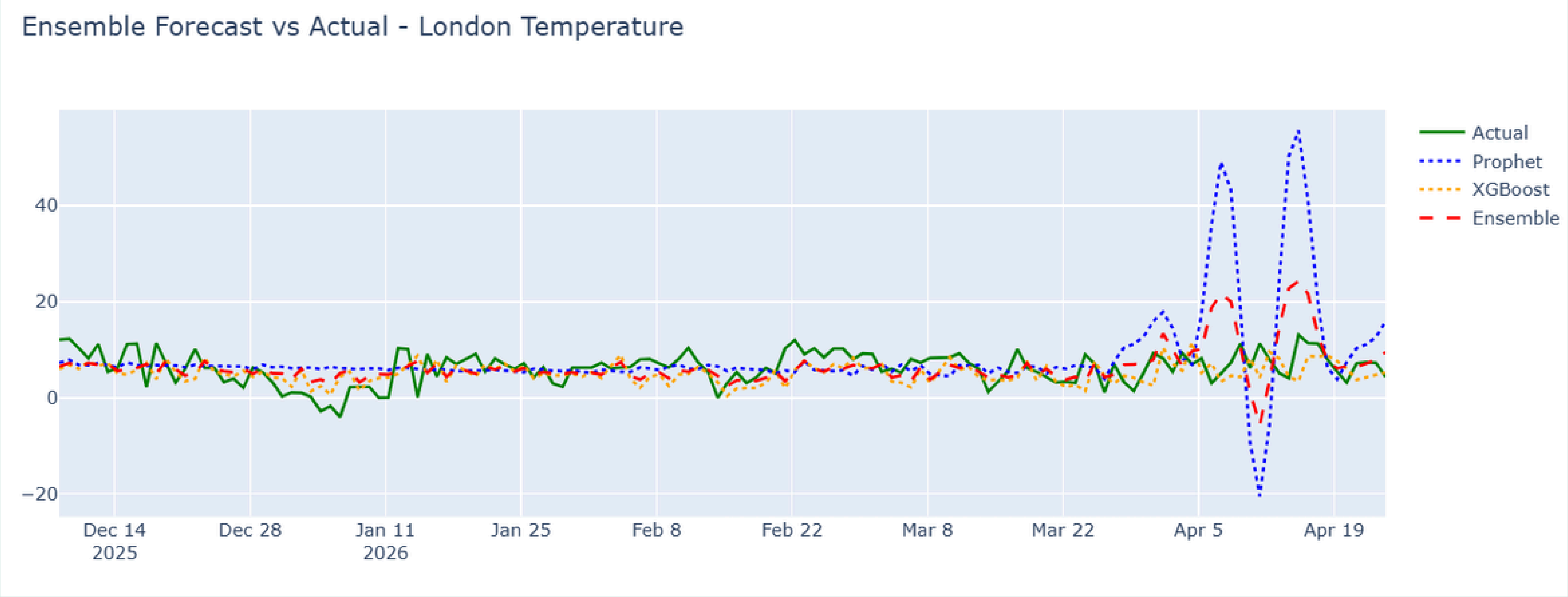


FORECASTING MODEL

Model	RMSE	MAE	R^2
Prophet	9.79	526	-7.55
XGBoost	3.57	2.91	-0.14
Ensemble	4.70	3.33	-0.97







KEY INSIGHTS AND CONCLUSION

- XGBoost best model — lag features captured hourly patterns effectively
- South Asia has worst air quality — temperature & humidity amplify PM2.5
- Anomaly detection found ~5% extreme weather events globally
- Temperature is #1 predictor of human thermal comfort
- Climate analysis shows clear inverse seasonality between hemispheres

THANK YOU